METRIC

MILITARY SPECIFICATION

BOLTS, SHOULDER, HEXAGON HEAD,
GENERAL SPECIFICATION FOR, METRIC

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of short thread and long thread hexagon head shoulder bolt with metric dimensions.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.

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DOD-B-63475/1 - Bolts, Shoulder, Hexagon Head, Long Thread, Metric.
DOD-B-63475/2 - Bolts, Shoulder, Hexagon Head, Short Thread, Metric.

STANDARDS

FEDERAL

FED-STD-H28/21 - Screw Thread Standards for Federal Services, Section 21, Metric Screw Threads.
FED-STD-66 - Steel: Chemical Composition and Hardenability.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1312 - Fasteners, Test Methods.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B18.3.3M-1979 - Hexagon Socket Head Shoulder Screws, Metric Series.

(Application for copies should be addressed to the American Society for Mechanical Engineers, United Engineering Center, 345 E. 47th Street, New York, NY 10017.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5306

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Bolts shall be made of alloy steel selected from FED-STD-66, of a composition consistent with mechanical properties specified in 3.4.

3.3 Design.

3.3.1 Threads. Bolt threads shall conform to the "M" profile and tolerance class 6g, after plating, as specified on the applicable specification sheet and as defined in FED-STD-H28/21.

3.3.2 Head taper. Bolt head taper (angle between any side of the head and the bolt axis) shall be not greater than 2 degrees.

3.3.3 Head position. The axis of the hexagon head shall be located at a true position with respect to the axis of the bolt (determined over a distance under the head equal to one shoulder diameter) within a tolerance zone of a diameter equal to 3 percent of the shoulder diameter for bolts of thread size M10 and smaller, and 6 percent for sizes larger than M10.

3.3.4 Concentricity. The shoulder diameter and thread pitch diameter shall be concentric within 0.10 mm, determined on the shoulder at an axial distance of 5 mm from the face of the shoulder.

3.3.5 Bearing surface. The bearing surface of the bolt head shall be perpendicular to the bolt axis within 2 degrees for shank lengths up to 100 mm, and within 1 degree for shank lengths over 100 mm.

3.4 Mechanical properties. Mechanical properties of bolts shall be as specified in table I and 3.4.1.

TABLE I. Mechanical properties.

Nominal size (M thread)	Minimum area (mm^2)		Shear strength (kN) min		Ultimate tensile 1/ load (kN) min
	Thread neck	Shoulder	Single 2/	Double 3/	
10	43.91	95.03	20.68	89.52	34.48
12	65.46	132.73	30.83	125.03	51.39
14	90.42	176.71	42.59	166.46	70.98
16	127.27	226.98	59.94	213.82	99.91
18	156.81	283.52	73.86	267.08	123.00
20	202.82	346.36	95.53	326.27	159.22
22	256.45	415.47	120.79	391.37	201.32
24	291.64	490.87	137.36	462.40	228.94
27	389.52	615.75	183.46	580.04	305.77
30	478.00	804.24	225.14	757.59	375.23
36	702.62	1134.11	330.93	1068.33	551.56

1/ Ultimate tensile strength of the bolt is based on a minimum ultimate tensile strength of 785 MPa for the material and the minimum thread neck area.
2/ Single shear strength is based on a stress of 60% of the minimum ultimate tensile strength (471 MPa) and the minimum thread neck area.
3/ Double shear strength is based on a stress of 60% of the minimum ultimate tensile strength (471 MPa) and twice the minimum shoulder area.

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3.4.1 Hardness. Bolts shall have a surface hardness of Rockwell 20-30HRC.

3.5 Decarburization. The depth of gross decarburization of thread surfaces shall be not greater than 0.015 mm. The depth of partial decarburization at the root shall be not greater than 10% of thread height, and at the crest shall be not greater than 25% of the thread height.

3.6 Protective finish. Bolts shall be cadmium plated in accordance with QQ-P-416, type II, class 3 (5.1 μm thick), or zinc plated in accordance with QQ-Z-325, type II, class 3 (5.1 μm thick), as specified (see 3.1).

3.7 Identification markings. Bolts shall be marked with the property class (see 6.4) and the manufacturer's name or registered trademark. Minimum height (size) of property class markings shall be 2.3 mm for M10 bolts, 3.2 mm for M12 and M14 bolts and 4.0 mm for M16 and larger bolts. Markings shall be located on the top of the head and may be raised or recessed. When raised, markings shall project not less than 0.1 mm for M14 and smaller bolts and 0.3 mm for M16 and larger bolts, above the surface of the head. The total height (head plus markings) shall not exceed the specified maximum head height plus 0.2 mm for M10 bolts, 0.3 mm for M12 and M14 bolts and 0.4 mm for M16 and larger bolts.

3.8 Workmanship. Finished bolts shall be uniform in quality and appearance. Surfaces shall be free of foreign matter. Bolt surfaces in the unplated condition shall be free of cracks, bursts, folds, tool marks, seams, voids on the bearing surface, nicks and gouges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections herein are classified as follows:

a. Materials inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that materials used in fabricating bolts are in accordance with 3.2, 3.4 and 3.6.

4.3.1 Protective finish. Sample bolts shall be inspected for minimum thickness and continuity of the plating in accordance with QQ-P-416 or QQ-Z-325, as applicable (see 3.1).

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A Examination	1/	4.5.1
Group B Mechanical properties Decarburization	 3.4 3.5	 4.5.2 4.5.3

1/ All requirements not covered by tests.

4.4.1 Inspection lot. An inspection lot shall consist of all bolts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.4.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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4.4.3 Sampling for group A inspection. Sampling shall be in accordance with MIL-STD-105 inspection level II with an acceptable quality level (AQL) of 1.5 percent defective for major defects. For minor defects, inspection level I with an AQL of 4.0 percent defective shall apply.

4.4.3.1 Classification of defects. Defects shall be classified as specified in table III.

TABLE III. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major		
101	Threads, not as specified (3.3.1).	Gage
102	Thread neck and fillet, under shoulder, not as specified (3.1).	SIE 1/
103	Width across flats, not as specified (3.1).	SIE
104	Shoulder diameter, not as specified (3.1).	SIE
Minor		
201	Overall length of bolt, not as specified (3.1).	SIE
202	Shoulder length, not as specified (3.1).	SIE
203	Other dimensions, not as specified (3.1).	SIE
204	Protective finish missing (3.6).	Visual
205	Neck and fillet under head, not as specified (3.1).	SIE
206	Head taper, not as specified (3.3.2).	SIE
207	Bearing surface - bolt axis perpendicularity, not as specified (3.3.3).	SIE
208	Workmanship, not as specified (3.8).	Visual
209	Packaging, not as specified (5.1).	Visual

1/ Standard inspection equipment

4.4.4 Sampling for group B inspection. Sampling shall be in accordance with MIL-STD-105 inspection level S-2 with an AQL of 1.5 percent defective.

4.4.4.1 Disposition of sample units. Bolts which have been subjected to group B inspection shall not be delivered on the contract.

4.4.5 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with PPP-H-1581.

4.5 Methods of inspection.

4.5.1 Examination. Sample bolts shall be thoroughly examined to determine conformance with this specification and applicable specification sheets with respect to all the requirements not covered by tests.

4.5.2 Mechanical properties. Sample bolts shall be tested for conformance to requirements of 3.4. Tensile strength and double shear strength shall be determined in accordance with MIL-STD-1312, test methods 8 and 13, respectively. Surface hardness shall be determined in accordance with MIL-STD-1312, test method 6.

4.5.3 Decarburization. When specified (see 6.2), sample bolts shall be tested for conformance with 3.5 in accordance with decarburization measurement methods of ANSI B18.3.3M.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Shoulder bolts are intended for use as pilots, guides, shafts, pivots and in other applications which require a bolt with a functional shoulder having closely held tolerances on diameter and concentricity.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Decarburization test, if required (see 4.5.3).
d. Level (degree) of protection, in accordance with PPP-H-1581, ordering data (see 5.1).

6.3 Definitions.

6.3.1 Metric terms and definitions. Metric terms used in this specification are defined in ASTM E 380, Standard for Metric Practice.

6.3.2 Quality assurance terms and definitions. Quality assurance terms used in this specification are defined in MIL-STD-109.

6.4 Property class. Property class, indicated by a numerical symbol, is the relationship between nominal tensile strength and nominal yield stress. The symbol consists of two numbers separated by a decimal marker (which has no mathematical significance), e.g., 8.8. The first number indicates 1/100 of the nominal tensile strength. The second number indicates 1/10 of the ratio, expressed as a percentage, between nominal yield stress and nominal tensile strength.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Army - AV, MI, AT
Navy - SH
Air Force - 11
DLA-IS
NSA-NS

User activities:
Army - CR, ME
Navy - AS, MC, OS

Preparing activity:
Army - AR

Agent:
DLA-IS

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